

Meher Chaitanya Pindiprolu, PhD

MACHINE LEARNING RESEARCHER

Efficient AI | LLM Optimization | Graph Machine Learning

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Profile

Machine Learning Researcher with a PhD in Network Science from ETH Zürich, specializing in compute- and memory-efficient AI. Leads research on LLM pretraining optimization, Tokenized Graph Transformers, long-range information flow, and scalable graph machine learning using optimization and numerical linear algebra. Combines research leadership with experience translating novel algorithms into production AI systems with measurable impact.

Work Experience

KTH Royal Institute of Technology

Stockholm, Sweden

WALLENBERG AI POSTDOCTORAL RESEARCHER

2025 – Present

- Define and lead research on efficient AI, originating projects in LLM pretraining optimization, tokenized graph transformers, long-range information flow, and algorithms for scalable graph learning.
- Drive end-to-end research from problem formulation to publication, including sparse Graph Transformers with up to 11% gains (NeurIPS 2026 Oral) and large-scale LLM optimization experiments.
- Provide scientific direction to PhD and Master's researchers, shaping methodology, experimental design, interpretation, and cross-university collaborations.

Coverstar

Stockholm, Sweden

AI CONSULTANT – RECOMMENDER SYSTEM ARCHITECT (part-time)

2025 – 2026

- Architected Coverstar's next-generation "For You" recommender, combining graph retrieval, session Transformers, cold-start methods, LLM-assisted content understanding, ranking, diversity, and safety.
- Identified that short-term engagement optimization reduced diversity and long-term retention, motivating a multi-objective ranking strategy balancing relevance, diversity, retention, latency, and scalability.
- Delivered a 14.3 percentage-point improvement in 30-day retention in A/B testing versus the NVIDIA Merlin production baseline.

ETH Zürich

Zürich, Switzerland

PHD RESEARCHER

2020 – 2024

- Introduced the first node-embedding framework connecting s-t min-cuts with contagion dynamics, improving node classification accuracy by up to 12%; also developed graph-Laplacian methods for link recommendation with provable guarantees.
- Designed the first scalable online algorithm for local clustering in multiplex networks, enabling efficient recovery of local structure without global graph exploration.
- Led SNSF-funded research from problem formulation through publication and mentored two Master's students on methodology, experiments, and interpretation.

IIIT Hyderabad

Hyderabad, India

HIGH PERFORMANCE COMPUTING RESEARCHER

2013 – 2016

- Developed high-performance parallel graph algorithms, achieving up to 4× speedups on large-scale graph problems.

NVIDIA

Bengaluru, India

SENIOR SOFTWARE ENGINEER

2012 – 2013

- Developed operating system kernel-level drivers for NVIDIA Tegra systems deployed in BMW automotive platforms.

Efficient LLM & Long-Range Information Flow Research

Graph Cascades: Mesoscopic Rewiring Improves Sparse Graph Attention	NeurIPS 2026 ORAL
Designed structured sparse-attention topologies for Graph Transformers that improve long-range information flow without dense attention, scaling to graphs with millions of nodes while improving predictive accuracy by up to 11%.	
What Does Muon Amplify?	Under review, 2026
Characterized when spectral-reweighting optimizers such as Muon amplify useful rather than noise-dominated gradient directions during LLM pretraining, validating the analysis across Pythia models up to 1B parameters.	
Schur Preconditioning for Long-Range Message Passing in GNNs	Under review, 2026
Applied preconditioning ideas from numerical linear algebra to mitigate over-squashing and improve long-range information flow in sparse neural architectures while approximately preserving the original interaction geometry.	
KV-Cache Compression in LLMs by Elimination, Not Eviction	Manuscript in preparation, 2026
Reformulated KV-cache compression as structured elimination that preserves interactions through removed tokens, retaining task-critical tokens in 88–100% of stress tests versus 12–50% for one-shot methods such as Top-K.	

Graph Learning & Numerical Algorithms Research

Fast Local Graph Clustering under Noisy Data	KDD 2027
Developed side-information-aware local graph clustering via shortest-path search, improving F1 by 6.4 points.	
Canonical Schur Trees for Approximate Cholesky in Laplacian Systems	Under review, 2026
Developed randomized sparse Schur-tree preconditioners for Laplacian systems underlying large-scale graph learning systems achieving 4.4–4.7× speedups over state-of-the-art solvers.	
Thresholded Local Hyperflow Diffusion	Under review, 2026
Engineered a hypergraph clustering method with provably local updates and early-stopping guarantees, improving median F1 with an average gain of 6.9 percentage points. 4	
Adjacency Search Embeddings	TMLR 2025
Designed node embeddings capturing higher-order structure, improving classification by up to 12.1% in GNNs.	

Education

ETH Zürich, Switzerland	2020 – 2024
PhD NETWORK SCIENCE	
Dissertation: Influence Processes on Networks	
Purdue University, USA	2016 – 2019
MSc COMPUTER SCIENCE — Focus: Numerical Optimization	

Core Expertise & Skills

AI: Efficient LLM Training KV-Cache Optimization Graph Machine Learning Long-Range Information Flow Graph Transformers Graph Neural Networks Recommender Systems
Methods: Numerical Linear Algebra Optimization Spectral Methods Graph Algorithms Randomized Algorithms
Programming & Frameworks: Python C++ C PyTorch PyTorch Geometric DGL scikit-learn NetworkX

Academic Service & Recognition

Awards:
Wallenberg AI Postdoctoral Fellowship Best Paper Award, HiPC Purdue–Qatar Fellowship AICTE Fellowship
Invited Talks:
Adjacency Search Embeddings, TU Wien, 2025 Influence Processes on Social Networks, UIUC, 2025
Academic Service:
Program Committee: AAAI Peer Review: NeurIPS, ICML, AISTATS